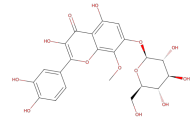
Eight peaks assemble a molecular decision

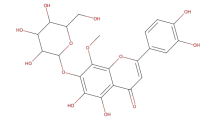
A frozen negative-ion case tested across all 256 peak combinations

UltraMS DreaMS



measured

$C_{22}H_{22}O_{13}$



same-formula isomer

a Sequential peak add-back

new peak retained peak unchanged spectrum

46 other peaks remain

8 selected peaks absent

0/8	1/8	2/8	3/8	4/8	5/8	6/8	7/8	8/8
	$C_7O_5H_3$ 165.99	$C_{15}O_8H_8$ 316.02	$C_6O_4H_9$ 138.00	$C_7O_5H_3$ 167.00	$C_{16}O_8H_{11}$ 331.04	$C_{14}O_6H_9$ 271.02	$C_{14}O_7H_9$ 287.02	$C_8O_6H_3$ 193.98
U #1 D #1	U #1 D #42	U #1 D #53	U #1 D #52	U #1 D #51	U #1 D #30	U #1 D #12	U #1 D #7	U #1 D #14

b All 256 combinations

UltraMS 256/256 at rank 1

DreaMS 51/256 at rank 1

